

LABORATORY WORK 1

APPLIED STATISTICS

Results Report

1. Dataset and data quality

The analysis was performed on the online-shopping session dataset. One row represents one recorded session. The variables describe session activity, customer information, subscription plan, purchase value and rating.

2. Data cleaning results

Before calculating descriptive statistics, the dataset was checked for duplicate records, missing values, sentinel values and impossible observations. Exact duplicate rows were treated as repeated records rather than additional independent sessions. Special placeholder values representing missing information were not included in numerical summaries. Negative session durations were treated as invalid because a session cannot have a negative length. Valid sessions with a missing rating were retained because the absence of a rating does not make the session itself invalid.

3. Main statistical findings

Session duration is expected to be right-skewed: most sessions are relatively short, while a smaller number of long sessions create a longer right tail. Consequently, the mean is more sensitive to extreme sessions than the median. The same issue is relevant to order value, where many sessions may have no purchase while completed orders create a positive right tail.

The analysis also distinguishes session-level and customer-level averages. A per-row average gives every recorded session equal weight, whereas a per-user average first summarizes each customer and therefore gives each customer equal weight. These two measures can differ when some customers generate many more sessions than others.

4. Rating variable

Rating is an ordinal variable. Its categories have a natural order, but the numerical distance between categories should not automatically be interpreted as equal. Therefore, the rating distribution, median, and proportions of low and high ratings are more informative than relying only on an arithmetic mean.

5. Conclusion

The main result of the laboratory work is that descriptive statistics depend strongly on data quality and measurement choices. Cleaning duplicates, invalid values and sentinels before analysis prevents distorted summaries. Robust statistics such as the median and IQR are useful when the distribution contains outliers or strong skewness.

HOW I USED AI IN THE LABORATORY WORK

1. Purpose of using AI

I used AI as an assistant to structure the laboratory work, understand statistical concepts, identify possible data-quality problems, and prepare the report. I used AI as support, not as a replacement for checking the actual data.

2. How I used AI

I asked AI to create predictions about the dataset before analysis.

I used AI to clarify statistical concepts and variable types.

I used AI to identify possible problems such as duplicates, missing values, and invalid observations.

I used AI to understand the difference between mean, median, IQR and standard deviation.

I used AI to organize my findings into a clear report.

3. Verification

AI suggestions were treated as hypotheses and were checked against the actual dataset. Numerical results and final conclusions were based on the data rather than on AI assumptions.

4. Reflection

AI helped me understand the statistical analysis and organize my work more efficiently. However, the laboratory work also showed me that AI-generated information should always be verified before being used as a final result.